CSE 115L - Lab 08: Written Answers

All code solutions are in main.c (each marked with its question number, e.g. //C8.1).

G8.1 Complete the switch

Blank	Answer
(1)	term
(2)	break
(3)	3
(4)	default

Input 1	Output
With the break	Spring
Without it	Spring, then Summer (control falls into case 2 and stops at its break)

G8.2 Which ladders can become a switch?

The ladder tests	Switch?	Why
mark >= 93, mark >= 90, ...	No	These are ranges; a case label is a constant, not a condition.
letter == 'A', letter == 'B', ...	Yes	Fixed char values, and char is an integral type.
gpa == 4.0, gpa == 3.7, ...	No	gpa is a double; a switch needs an integral expression.
month == 1, month == 2, ...	Yes	Fixed int values.

E8.1 Predict the cascade (all breaks removed)

Input	Lines predicted	Lines actually printed
A	5	Strong work. Keep the same study pattern. A pass. Review the topics you lost marks on. At risk. See the course faculty this week. Not passed. The course must be repeated. A is not a grade on the NSU scale.
C	4	A pass. Review the topics you lost marks on. At risk. See the course faculty this week. Not passed. The course must be repeated. C is not a grade on the NSU scale.
F	2	Not passed. The course must be repeated. F is not a grade on the NSU scale.
K	1	K is not a grade on the NSU scale.

Entering at a label, control runs every statement below it, including default, until the closing brace.

M8.2 Five paths tested

Input (a b op)	Output	Path
6 3 +	6.00 + 3.00 = 9.00	case '+'
6 3 -	6.00 - 3.00 = 3.00	case '-'
6 3 *	6.00 * 3.00 = 18.00	case '*'
6 3 /	6.00 / 3.00 = 2.00	case '/', divisor non-zero

Input (a b op)	Output	Path
6 0 /	Cannot divide by zero.	case '/', zero refused
6 3 ^	Unknown operator '^'.	default

The five paths are +, -, *, / (with its zero check) and default; the zero check is the sixth row because it is the refusal branch inside /.

M8.3 Every break removed, input 3 4 +

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3.00 + 4.00 = 7.00
3.00 - 4.00 = -1.00
3.00 * 4.00 = 12.00
3.00 / 4.00 = 0.75
Unknown operator '+'.
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Why the last line is that one: with no break, control entered at case '+' and ran straight through every later case to the closing brace, so it also ran default, the final statement in the switch, which reports '+' as unknown. Breaks restored in the M8.2 version.

C8.1 N % 13

N % 13 gives a remainder from 0 to 12, which is 13 possible values for only 12 months. The values 1 to 12 are valid months, but 0 (when N is a multiple of 13) is not a month, so the validation (month < 1 || month > 12) must catch it and print '0 is not a month number.' before the switch runs. N is never negative, so only 0 can slip through; the validation also covers negative and too-large numbers for other inputs.

C8.2 Find the four defects

Type the listing with the printf under the if indented one level further (as printed in the manual); that indentation is what lets gcc warn about defect 3.

#	What is wrong	Compiler	Corrected line
1	switch (gpa): the controlling expression is a double.	**Stopped** (error: switch quantity not an integer)	switch ((int)gpa) {
2	case 3: appears twice (line 17).	**Stopped** (error: duplicate case value)	case 2:
3	if (credits >= 3); - the stray semicolon is the whole if body, so 'A full course.' always prints.	**Warned** (-Wmisleading-indentation)	if (credits >= 3)
4	No break after case 4's printf, so input 4 prints 'Highest grade.' and 'Good standing.'	**Said nothing**	add break; after printf("Highest grade.\n");

Fix order: defects 1 and 2 first (both reported together), rebuild; then the warning for 3 appears; defect 4 is found only by running with input '4 3' and reading the output.

Which single defect makes a switch on a grade point a bad idea? Defect 1. A grade point is a double (3.7, 3.3, ...), and a switch needs an integral controlling expression with integer constant labels. The (int) cast only makes it compile: it truncates 3.7 to 3, so distinct grade points collapse together. A ladder is the right tool.

Layout change that hides the warning while the defect stays: indent the printf to the same level as the if (flush with it), i.e. 'if (credits >= 3);' and 'printf(...);' in the same column. The semicolon still ends the if, so the printf still always runs; only the misleading layout, and therefore the warning, is gone.